

# **CIRA: Cascaded Implicit Recurrent Attention**

**A semantic-to-contextual language model with entropy-regularized fixed-point attention.**

**Gary Nan Tie, Sep 11, 2026**

## **Abstract**

We introduce **Cascaded Implicit Recurrent Attention (CIRA)**, a language-model architecture that composes two implicit recurrent state machines to couple semantic representation with contextual token prediction. The first implicit recurrent model learns a state trajectory together with equilibrium fixed points that serve as structured key representations, while its output is trained in a word-embedding space to encourage semantic coherence. A second implicit recurrent model operates on the resulting semantic states and learns a contextual state trajectory together with its own equilibrium fixed points serving as attention values. Rather than treating queries, keys, and values as unconstrained collections of learned vectors, CIRA derives them from the trajectories and fixed-point dynamics of the two recurrent systems.

An attention-based output function is constructed by stacking feedforward attention layers over the learned key and value states. Causal attention produces contextual representations, followed by a final attention operation that selects a learned fixed-point value associated with the predicted token continuation. Training minimizes token-level perplexity jointly with a Rényi-entropy regularization term on the continuation attention distribution. The resulting objective encourages accurate prediction while controlling the concentration of the model's attention over its learned contextual states.

CIRA therefore provides a structured alternative to conventional transformer attention: semantic state formation precedes contextual state formation, and attention operates over dynamically generated equilibrium representations rather than independently parameterized key-value memories. The architecture offers an interpretable state-machine decomposition in which semantic and contextual dynamics, fixed-point representations, and token prediction can be examined separately. This provides a framework for investigating whether structured implicit dynamics and entropy-regularized attention can improve the semantic coherence, efficiency, and auditability of language models.

## Introduction

Compose implicitRNNs {1} and {2} as follows:

{1}

$g_0 := 0, k_0 := 0, i \in [m]$

input  $u_i := [t_i \ g_{i-1}]'$

$k_i = \phi( A k_i + B u_i )$  (equilibrium equation)

$g_i = S(g_{i-1}, t_i) := C k_i + D u_i$  (prediction equation)

(where  $A := \Gamma - \text{diag}(|\Gamma| \mathbf{1}) + \gamma \text{Id}$ ,  $0 < \gamma < 1$ ,

ensures  $A$  well-posed for  $\phi$ , that is,

the fixed point equation has a solution and it is unique).

Let  $o_i = O(g_i) := \text{FFN}_\theta(g_i)$  where  $\text{FFN}_\theta$  is a feedforward network with parameters  $\theta$ .

State machine (S,O) training problem:

Learn matrices  $A, B, C, D$  and fixed points  $k_i$  and parameters  $\theta$  to minimize:

Sum over  $i \in [m]$   $\{1 - \text{Cos}(o_i, \text{Emb}(t_i))\}$

where  $\text{Emb}$  is a word embedding.

{2}

$h_0 := 0, v_0 := 0$

input  $w_i := [g_i \ h_{i-1}]'$

$v_i = \psi( E v_i + F w_i )$  (equilibrium equation)

$h_i = S'(h_{i-1}, g_i) := G v_i + H w_i$  (prediction equation)

(where  $E$  is well-posed for  $\psi$ ).

We construct output function  $O'$  corresponding to transition function  $S'$ , from attention layers given the  $k_i$  and  $g_i$  from {1} as follows:

Attention Layer

Input  $h_i$ .

For  $i \in [m]$ , define probabilities

$\alpha_{i,j} = \text{softmax over } j \in [i] \text{ score}(h_i, k_j)$

where  $\text{score}$  is a similarity function.

Let  $h^i = \text{Sum over } j \in [i] \alpha_{i,j} v_j$  and define update  $h'_i = \text{FFN}_\theta(h^i)$ .

Output  $h'_i$ .

Note that  $k_i$  and  $g_i$  are from  $\{1\}$  and that  $v_i$  and  $h_i$  are from  $\{2\}$ .

Stack  $L$  attention layers with parameters  $\Theta = \{\theta_1, \dots, \theta_L\}$ , composing updates  $\text{FFN}_\theta$  and now let  $h'_i$  denote the final  $L$ -th update. After  $L$  attention layers, for  $i \in [m]$  define probabilities  $\beta_{i,j} = \text{softmax over } j \in [i] \text{ score}(h'_i, v_j)$  and let  $j^*_i = \arg \max \text{ over } j \in [i] \beta_{i,j}$ .

Define token continuation  $o'_{i+1} = O'(h_i) := t_{(j^*_i)}$ .

This is how attention layers construct output  $O'$  corresponding to transition  $S'$  in  $\{2\}$ .

Define average Renyi  $\alpha$ -entropy of  $\{t_i\}$  by:

$$E(\{t_i\}) = (1/m) \sum \text{ over } i \in [m] H_i$$

where  $H_i = H_\alpha(\{\beta_{i,j}\})$  and  $H_\alpha$  is Renyi  $\alpha$ -entropy.

Token continuation via  $(S', O')$  training problem:

Let  $\lambda > 0$ , learn matrices  $E, F, G, H$  and fixed points  $v_i$  in  $\{2\}$  and feedforward layer parameters  $\Theta$  to minimize (\*):

$$\text{Perplexity}(o'_2, \dots, o'_{m+1}) + \lambda E(\{t_i\})$$

Summary:

In the cascade of implicit models, we first learn the state machine  $(S, O)$  from  $\{1\}$ , in particular keys  $k_i$ , and then from  $\{2\}$  we learn  $(S', O')$ , in particular values  $v_i$  and token contextual representations  $h_i$ , thereby creating a new attention mechanism with regularized objective (\*).

## Appendix

$$\begin{array}{c}
 u_i = \begin{pmatrix} t_i \\ g_{i-1} \end{pmatrix} \\
 \downarrow \\
 \boxed{
 \begin{array}{l}
 k_i = \varphi(Ak_i + Bu_i) \\
 g_i = Ck_i + Du_i
 \end{array}
 } \quad (1) \\
 \leftarrow \\
 w_i = \begin{pmatrix} g_i \\ h_{i-1} \end{pmatrix} \\
 \downarrow \\
 \boxed{
 \begin{array}{l}
 v_i = \gamma(Ev_i + Fw_i) \\
 h_i = Gv_i + Hw_i
 \end{array}
 } \quad (2)
 \end{array}$$

Composition of implicit recurrent models.

From (1) we will define a state machine  $(\mathcal{S}, \mathcal{O})$ :

Input: target tokens  $t_1, \dots, t_m \in \mathbb{R}^P$

Initial state:  $g_0 = 0 \in \mathbb{R}^q$

Next state:  $g_i = \mathcal{S}(g_{i-1}, t_i) \in \mathbb{R}^q, i \in [m]$

Output:  $o_i = \mathcal{O}(g_i) \in \mathbb{R}^P, i \in [m]$

where  $\mathcal{S}$  is a transition function and

$\mathcal{O}$  is an output function as follows:

—

Let  $\Theta = \{(W_1, b_1), (W_2, b_2)\}$  be parameters

for a feedforward network defined by

$$\text{FFN}_{\Theta}(g_i) \triangleq \text{ReLU}(g_i W_1 + b_1) W_2 + b_2$$

—

Let  $\text{Emb}: \mathbb{R}^P \rightarrow \mathbb{R}^{P'}$  be a word embedding, like word2vec.

—

$$\text{Let } \text{Cos}(a, b) \triangleq \frac{\langle a, b \rangle}{\|a\| \|b\|}$$

implicit RNN (1):

$$g_0 = 0 \in \mathbb{R}^q, \quad k_0 = 0 \in \mathbb{R}^q$$

$$u_i \triangleq \begin{pmatrix} t_i \\ s_{i-1} \end{pmatrix} \in \mathbb{R}^r, \quad r = p+q$$

$$k_i = \varphi(Ak_i + Bu_i) \in \mathbb{R}^q$$

$$g_i = \mathcal{S}(g_{i-1}, t_i) \triangleq Ck_i + Du_i \in \mathbb{R}^q$$

where  $A \triangleq \tilde{J} - \text{diag}(|\tilde{J}| \mathbf{1}_q) + \gamma \text{Id}_q$ ,  $0 < \gamma < 1$ ,

ensures  $A$  is well-posed for  $\varphi$ , that is

the fixed point equation has a solution and

it is unique.

—

$$o_i = \mathcal{O}(g_i) \triangleq \text{FFN}_{\theta}(g_i)$$

$(\mathcal{S}, \mathcal{O})$  training problem

Learn matrices  $A, B, C, D$  and fixed points  $k_i$  and parameters  $\theta$

to minimize:  $\sum_{i=1}^m 1 - \text{Cos}(o_i, \text{Emb}(t_i))$

So  $(\mathcal{S}, \mathcal{O})$  simulates word embedding  $\text{Emb}$ .

Similarly, (2) defines another state machine  $(\mathcal{S}', \mathcal{O}')$ :

$$h_0 = 0 \in \mathbb{R}^q, \quad v_0 = 0 \in \mathbb{R}^q$$

$$w_i = \begin{pmatrix} g_i \\ h_{i-1} \end{pmatrix} \in \mathbb{R}^r, \quad r = p+q$$

$$v_i = \mathcal{U}(E v_i + F w_i)$$

$$h_i = G v_i + H w_i =: \mathcal{S}'(h_{i-1}, g_i)$$

where  $E$  is well-posed for  $\mathcal{U}$ .

Attention layers will construct output function  $\mathcal{O}'$ .

Attention layer (given  $\{g_i\}$  from (1) )

Input  $h_i$ . Let score be similarity function eg a kernel.

For  $i \in [m]$ , define probabilities  $\alpha_{ij} = \frac{\text{score}(h_i, k_j)}{\sum_{j \in [i]} \text{score}(h_i, k_j)}$

Let  $\hat{h}_i = \sum_{j=1}^i \alpha_{ij} v_j$  and define update  $h'_i = \text{FFN}_\theta(\hat{h}_i)$  ( $\dim h'_i = \dim k_j$ )

Output  $h'_i$

✗

Note that  $k_i, g_i$  are from (1) and  $v_i, h_i$  are from (2).

Stack  $L$  attention layers with parameters

$\Theta \triangleq \{\Theta_1, \dots, \Theta_L\}$ , composing updates  $\text{FFN}_{\Theta_i}$

and now let  $h'_i$  denote the final  $L$ -th update.

After  $L$  layers, for  $i \in [m]$  define probabilities

$$\beta_{ij} = \text{softmax}_{j \in [i]} \text{score}(h'_i, v_j) \text{ and}$$

$$\text{let } j_i^\beta = \arg \max_{j \in [i]} \beta_{ij}.$$

Define token continuation  $o'_{i+1} = \Theta'(h_i) \triangleq t_{j_i^\beta}$

This how attention layers construct output  $\Theta'$

→

Define the average Renyi  $\alpha$ -entropy of  $\{t_1, \dots, t_m\}$  by:

$$E(\{t_i\}_{i=1}^m) \triangleq \frac{1}{m} \sum_{i=1}^m H_\alpha^i$$

where  $H_\alpha^i \triangleq H_\alpha(\{\beta_{ij}\}_{j=1}^i)$  and  $H_\alpha$  is Renyi  $\alpha$ -entropy

→

Token continuation  $t_1, t_2, \dots, t_i, o'_{i+1}$  via:

$(\mathcal{S}', \mathcal{O}')$  training problem

Let  $\lambda > 0$ , learn matrices  $E, F, G, H$  and fixed points

$v_i$  in (2), and feedforward layer parameters  $\Theta$

to minimize:  $[ \text{Perplexity}(o'_2, \dots, o'_{m+1}) + \lambda E(\{t_i\}_{i=1}^m) ] \quad (*)$

→

Summary

In the cascade of implicit models, we first learn the state machine  $(\mathcal{S}, \mathcal{O})$  from (1), in particular keys  $k_i$ , and then from (2) we learn  $(\mathcal{S}', \mathcal{O}')$ , in particular values  $v_i$  and token contextual representations  $h_i$ , thereby creating a new attention mechanism with regularized objective  $(*)$ .

## References

[1] 'Alice Through the Word Embedding'

Gary Nan Tie, Jul 2, 2026

DOI: 10.13140/RG.2.2.34901.79841

[https://www.researchgate.net/publication/408460384\\_Alice\\_Through\\_the\\_Word\\_Embedding?channel=doi&linkId=6a4921a95ec65810298eecaf&showFulltext=true](https://www.researchgate.net/publication/408460384_Alice_Through_the_Word_Embedding?channel=doi&linkId=6a4921a95ec65810298eecaf&showFulltext=true)

[2] 'Alice in Parallel'

Gary Nan Tie, Jul 18, 2026

DOI: 10.13140/RG.2.2.18755.36641/1

[https://www.researchgate.net/publication/410626715\\_Alice\\_in\\_Parallel?channel=doi&linkId=6a64f3e2675a850148c9fa9a&showFulltext=true](https://www.researchgate.net/publication/410626715_Alice_in_Parallel?channel=doi&linkId=6a64f3e2675a850148c9fa9a&showFulltext=true)

[3] 'Learning Textual Entailment and Abduction'

Gary Nan Tie, Aug 13, 2026

DOI: 10.13140/RG.2.2.28712.53765

[https://www.researchgate.net/publication/412266441\\_Entailment\\_Abduction?channel=doi&linkId=6a7f76d98f3f8834d8f2f889&showFulltext=true](https://www.researchgate.net/publication/412266441_Entailment_Abduction?channel=doi&linkId=6a7f76d98f3f8834d8f2f889&showFulltext=true)

[4] 'Implicit Meta-Attention'

Gary Nan Tie, Aug 28, 2026

DOI: 10.13140/RG.2.2.26345.86886

[https://www.researchgate.net/publication/413720030\\_Implicit\\_Meta-Attention?channel=doi&linkId=6a91c4c47e8edd1da26fa96a&showFulltext=true](https://www.researchgate.net/publication/413720030_Implicit_Meta-Attention?channel=doi&linkId=6a91c4c47e8edd1da26fa96a&showFulltext=true)

[5] 'The Extrapolation Power of Implicit Models'

Juliette Decugis, Alicia Y. Tsai,

Max Emerling, Ashwin Ganesh, Laurent El Ghaoui

arXiv:2407.14430v1 [cs.LG] 19 Jul 2024